

Data Handling: Import, Cleaning and Visualisation

Exercise 4: Data Import and Data Preparation/Manipulation

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Exercise A

Read Section 11.-11.4 in <https://r4ds.had.co.nz/data-import.html> and solve the problem posed in Section 11.4.2 (“Problems”) by going step-by-step through the instructions. Hint: You will need to have the **readr** package installed and loaded (should be also installed/loaded with **tidyverse**).

Exercise B

1. Download the file `airquality_data.xlsx` posted on Canvas and save it in your `r_course/data` folder.
2. The file is a dataset that contains the New York air quality measurements. Familiarize with the data using `print`. Do you think the dataset has problems? (Hint: look at the **type** of each column).
3. Fix the problem identified in point 2. Save the new dataset as `airquality_data_fixed.xlsx` using `write_xlsx` from `tidyverse/writexl`.
4. Write your solution down in an R-Script and document in this script why certain problems occur and how you solved them (using comments).

Exercise C

1. Visit the EU Open Data Portal on COVID cases. (Do you remember this website from Exercise Session 2?)
2. Download the “COVID-19 Cases Worldwide” data in JSON format. However, do not download these data manually (i.e., do not click-and-save). Instead, load the data directly into RStudio from the web. Hint 1: consider the `jsonlite` package. Hint2: the direct link to the JSON file (<https://opendata.ecdc.europa.eu/covid19/casedistribution/json/>) may be helpful.
3. Reformat the JSON data that you have imported into RStudio to a rectangular format (such that each record is represented as a row and each field as a column).

Mock Exam Questions

Part 1: TRUE/FALSE Questions

For each of the following statements, decide whether the statement is TRUE (T) or FALSE (F).

1. **for-loops** are typically used in a situation where a sequence of commands needs to be executed for a well-known number of times.

Part 2: Multiple Choice Type A

For each of the following questions/statements, decide which of the possible responses are correct. NOTE: Either ONE OR SEVERAL of the response statements can be correct. In order to get a point, all correct

response statements need to be marked!

1. After executing the following lines of R code, which value is stored in `a[3]`?

```
a <- c(1:3)
```

```
a <- a*3
```

- A) 27
 - B) 3
 - C) 1:3
 - D) 9
-

Part 3: Multiple Choice Type B (only one correct!)

For each of the following questions/statements, decide which of the possible responses is correct. NOTE: For each question only ONE of the possible responses is correct. Select ONLY ONE of the response statements per question!

1. The following R code was executed on the command line and the corresponding error occurred.

```
read_csv("data/file.csv")
```

```
Error: 'data/file.csv' does not exist in current working directory ('/home/user/my_projects').
```

Which of the following statements is correct in this context (assuming `file.csv` actually is stored somewhere on this computer's harddrive)?

- A) Given the current working directory of this R session, the relative path “data/file.csv” is wrong and `read_csv()` therefore does not find the file.
- B) `file.csv` is generally not a valid file name.
- C) `read_csv()` is not the right R function to import `file.csv`.
- D) `read_csv()` fails to parse `file.csv` because the wrong column delimiter is used (commas).